

Personal Programming Project Report

Title: Neele, a new version of Wordle

Provide a descriptive title for your project.

1. **(5pts) Honor Code and LLM Usage for this Report.**

I have neither given nor received unauthorized assistance on this assignment.

During the preparation of this assignment, Neel Ramachandran Claude in debugging and assisting to help create and run the code After using this tool. I also used it to explain difficult concepts. , I reviewed and edited the content as needed to ensure its accuracy and take full responsibility for the content in relation to grading.

2. **(15pts) Learning Objectives:**

List the learning objectives from your proposal. In your own words explain whether you met those objectives and how (50-100 words each objective). Also describe if you learned something different than expected or anything additional.

The learning objectives from my proposal were conceptual understanding and skill development. My conceptual understanding learning objective was to learn and understand how word-based games like Wordle work (like the letter and position matching systems). I met this objective by creating a word game that gave the user six guesses to guess a five letter word based on a category chosen by the user.

My skill development objective was to improve my coding skills by building a fully functional wordle-style game. I wanted to include different Wordle categories too. Like someone could choose the category food, and they would only get five letter Wordles on food. I completed this learning objective by gaining more experience in JavaScript.

3. **(15pts) Timeline:**

Outline how you spent time on your project. Break down the time into specific tasks or milestones. Here is an adjustable schedule to get you started. Actual Details should be 50-100 words each and should compare or reflect on differences from your proposal.

Time	Task	Expected Details from Proposal	Actual Details
Hour 1-2	Research and gather resources	Research how Wordle works and figure out all the rules. Figure out which resources and large	I researched many articles on the origins of Wordle and how it worked. I also looked up and researched which LLMs would

Time	Task	Expected Details from Proposal	Actual Details
		language models I will use to do my project.	be the most helpful for me in making this project.
Hour 3-4	Design the project structure and plan	Make a draft of what the project will look like and a draft of the app. A draft of successfully solving wordle and winning the game, and a draft of unsuccessfully solving wordle and losing the game.	I created some rough sketches on what the project would look like. I also asked my friends for any suggestions or tips on anything I could add to my final design. In my sketches I made sure to include the winning and losing screens.
Hour 5-6	Start coding the basic functionalities	Start implementing the algorithms and checks I will use to create my Wordle app. Create the final product design and implement it. Write all the code and ensure everything is correct.	Started coding and learning the basic rules and layout of wordle. Spent a lot of time trying to create the layout and the GitHub as I kept running into issues. Organizing everything into the file hierarchy was also a challenge, but it was very helpful.
Hour 7-8	Test and debug the initial version	Go through and debug and check all of the code I have written. Fix all of the possible errors in the code. Make sure the UI design is polished and doesn't have any bugs.	Got stuck many times and used the assistance of LLMs (especially Claude) to nudge me in the right direction and to point out my errors. I also spent a lot of time trying to google solutions and google problems in my code to try and find ways to fix them.
Hour 9-10	Refine and add advanced features	Play the game many times to check and see if anything is wrong. Change elements of the UI to make the game more	Upgraded the UI designs to have different screens, different colors, and win and loss screens. Shared the game with my roommates to get their opinions on it. I made sure I listened to what they liked and

Time	Task	Expected Details from Proposal	Actual Details
		appealing. Share the game with others to get their opinion.	what they didn't like and thought about what I could change to make the game better.
Additional			

4. **(55pts) Final Product Description:**

Include your proposed MVP, Target, and Reach versions.

i. Minimum Viable Product (MVP):

A game like Wordle, where someone has 6 guesses to successfully guess a word. Correct letters will either be marked as green or yellow. Incorrect letters will be marked as grey. Target audience is anyone who enjoys the game of wordle.

ii. Target Product:

The game of wordle. A program which when you run launches the game. You will be able to guess a word within six tries. You will be given hints when you input a correct letter.

iii. Reach Version:

The same product as the target product, except a streak counter that counts your consecutive wins, a win/loss history, and the ability to make your own Wordles and your own categories.

iv. (20pts) Description of final product including target audience, user story, problem statement, key features, technical details and technologies used. (100 – 150 words)

The final product consists of a screen with 6 categories (food, sports, animals, countries, family, and random). Clicking a category will give you a five letter word based on the category that the user has to solve. Correct letters and placements will be colored green and correct letters but incorrect placements will be colored as yellow. If the user successfully guesses the wordle, they will be shown a win screen and if they do not guess the wordle, they will see a loss screen with the answer. A button to play again is shown on both screens. I used Claude AI to help assist with writing the code and running everything.

v. (20pts) Provide a YouTube link to your video demonstration (1–2 minutes, narrated). **Important Note:** Do not upload your video file directly. Instead, upload your video to YouTube and include the video link clearly here in your report. The level of difficulty and detail of the project should be reasonable for

10 hours of work with LLM support. The project should not be something an LLM can solve without significant effort by the developer.
(Be sure to have someone else test that your link is working.)

<https://youtu.be/AhFoQBbOGS8>

- vi. (15pts) Any input files, coding files, and test files should be uploaded. Provide a list here of file names and purposes, or any links to live sites or artifacts. Remember code should also be commented. A README file should be created and uploaded so that we have the option to follow your instructions to run your project.

GITHUB LINK:

<https://code.vt.edu/neel/wordle-game>

I know it is public, the GTA's GitHub username wasn't listed anywhere, so I couldn't give them access. I made it public to ensure that it could be graded.

Project Repository & Code Submission Details:

Project Repository (code.vt: You <https://code.vt.edu/> or (Git-hub only if you are part of virtual global collaboration)): r repository should be well-organized, documented, and easy to navigate. At a minimum, include the following structure:

- **code/** – All source code files for your project (organized by component or module if applicable).
- **data/** – Any input files, datasets, or configuration files used by your program.
- **tests/** – Test scripts or files demonstrating how your code was verified.
- **docs/** – Supporting materials such as screenshots, reports, or documentation.
- **report/** – This final report document.
- **README.md** – A detailed file describing:
 - Project overview and purpose
 - Video link of your project
 - Installation and setup instructions
 - How to run the program and reproduce results
 - Technologies or libraries used
 - Author(s) and contribution summary

Required:

- Maintain a logical directory structure, do not store all files at the root level.
- Include comments in your code to explain logic and design decisions.
- Keep your repository **private** until grades are released, then you may make it public.

Share access with the following personnel (Add them as collaborators):

GTA Name	Section	Professor
Mona Moghadampanah	83484	P. Sullivan
Yue Shen	83485	O. Emebo
Abdullah Al Noman	83486	O. Emebo
Suraj Vishwanath	83487	P. Sullivan
Juno Bartsch	91578	S. Nizamani

5. **(10pts) Consultation and Use of LLMs:**

Each student must create a unique project but is allowed to consult with other people and use Large Language Models (LLMs). Describe how you incorporated these resources into your project:

I incorporated LLMs to help me fix my code when it didn't work. I also used LLMs to give me a good basic outline for the project. I also used LLMs to debug and correct my code and to add comments to better help me understand my code. I also used the help of LLMs to upload everything to gitHub.

- **Consultation Description:**

Describe how you ended up seeking advice or feedback from peers, mentors, or online communities.

I got feedback from my roommates and friends on the game. I asked them what they liked, what they didn't like and what they wanted to change.

- **Use of LLMs:**

Explain how you ended up utilizing LLMs to assist with coding, debugging, learning technologies and concepts, or generating ideas.

I used LLMs for coding, debugging, learning, and generating ideas. I used LLMs to assist with coding when I got stuck or didn't know what to do. I used LLMs to assist with debugging code that didn't work. I used LLMs to explain difficult concepts to me. Finally, I used LLMs to give me suggestions to things I could add to my project.